

Animal Herd Detection Web App (Flask + YOLO)

Introduction

This project is a Flask-based web application that detects animals in an uploaded image using the YOLOv8 deep learning model. If animals are detected, the system retrieves a location using the NASA Earth API and displays that location on a map using the Folium library.

Libraries Used

Flask: Used to create the web application and handle user requests.

OpenCV (cv2): Used for image processing and saving detection results.

Ultralytics YOLO: Used for object detection to identify animals in images.

Requests: Used to call the NASA API to get location data.

Folium: Used to generate an interactive map with the detected location.

Main Functions

`get_location_from_nasa()`: Retrieves latitude and longitude using NASA Earth imagery API.

`detect_animals()`: Runs the YOLO model on the uploaded image and checks if animals are detected.

`create_map()`: Generates a map with a marker showing where animals were detected.

`index()`: Flask route that handles image upload, detection, and displaying results.

Working of the System

The user uploads an image through the web interface. The YOLO model processes the image and detects animals. If an animal is found, the system obtains geographic coordinates from the NASA API and generates a map showing the detection location. The result image with detection boxes and the map are then displayed on the webpage.

Conclusion

This project demonstrates how computer vision and web technologies can be combined to detect animals automatically and visualize their location on a map.